E-COMMERCE SALES ANALYSIS AND CUSTOMER INSIGHTS

A Data Analytics Case Study Using Python, SQL, and Power BI

Project Domain: E-Commerce | Retail Analytics | Business Intelligence
Project Type: Data Analysis and Business Decision-Making
Tools and Technologies: Python, SQL, Microsoft Power BI, Excel
Key Areas: Sales Performance, Customer Segmentation, Product Analysis, Revenue Trends, Business Optimization

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1. ABSTRACT

The rapid growth of e-commerce has transformed how businesses sell products and interact with customers. Online retailers generate substantial amounts of data through customer orders, transactions, product views, payments, and returns.

Analyzing this data can help businesses understand customer purchasing behavior, identify profitable products, evaluate sales performance, and improve operational efficiency.

This case study examines a hypothetical online retail business experiencing challenges in understanding its sales trends, customer retention, product performance, and revenue opportunities.

The objective is to develop a data-driven analytical framework that converts raw transactional data into meaningful business insights.

The proposed analysis focuses on sales performance, monthly revenue trends, product category performance, customer purchasing patterns, average order value, and repeat customer behavior.

Python is used for data cleaning and exploratory data analysis, SQL for querying and aggregating transactional data, and Power BI for interactive dashboard development.

The study demonstrates how data analytics can support business decisions, improve visibility into key performance indicators, and help e-commerce businesses identify opportunities for sustainable growth.

Note: This is a hypothetical case study. The business scenario, dataset structure, and recommendations are illustrative. No real company's transaction data has been analyzed, and no numerical business results are claimed.

2. INTRODUCTION

E-commerce businesses operate in a competitive environment where understanding customers and responding to market changes are essential for success.

Online retailers must manage multiple business activities, including product listings, inventory, pricing, order processing, delivery, customer service, and marketing.

Each activity generates data that can be analyzed to evaluate business performance.

However, collecting data alone does not guarantee better decisions. Businesses need effective analytical methods to identify patterns, measure performance, and understand the factors influencing their results.

Data analytics enables businesses to answer important questions, such as:

- Which products generate the highest revenue?
- Which categories contribute most to overall sales?
- During which months does the business perform best?
- How much does the average customer spend per order?
- What proportion of customers make repeat purchases?
- Which customer groups contribute the most revenue?
- Where can the business improve its sales and customer retention strategies?

This case study explores how these questions can be addressed using a structured data analytics workflow.

The goal is to transform transactional data into actionable insights that support sales optimization, customer engagement, and business growth.

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3. BUSINESS SCENARIO

Consider a hypothetical online retail company called **ShopSphere**.

ShopSphere sells products across several categories, including electronics, clothing, home essentials, beauty products, and accessories.

The company receives orders through its website and mobile application.

Despite having a growing customer base, management faces several challenges.

3.1 Business Challenges

1. Limited Sales Visibility

Management lacks a consolidated view of revenue, order volume, and sales trends.

2. Unclear Product Performance

The company cannot easily identify which products and categories generate the greatest revenue or which products sell frequently but contribute relatively little revenue.

3. Customer Retention Challenges

The business has limited visibility into repeat purchasing behavior and customer lifetime value.

4. Marketing Inefficiency

Marketing campaigns are not consistently evaluated against customer acquisition, conversion, and repeat purchasing metrics.

5. Inventory Planning Difficulties

The company needs better information about product demand to reduce the risk of stockouts and excess inventory.

6. Inconsistent Reporting

Different departments use separate spreadsheets, making it difficult to maintain a consistent understanding of business performance.

3.2 Business Objective

The primary objective is to develop a centralized analytical framework that enables management to monitor sales performance, understand customer behavior, evaluate product categories, and make more informed business decisions.

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4. PROBLEM STATEMENT

The hypothetical e-commerce business generates transactional data but lacks a systematic process for converting it into actionable business intelligence.

Without reliable analysis, management may struggle to identify sales opportunities, retain valuable customers, allocate marketing budgets effectively, and optimize inventory.

The project aims to analyze e-commerce transaction data and develop a dashboard that answers key business questions.

The central question is:

How can data analytics help an e-commerce business understand sales performance, improve customer retention, and make better commercial decisions?

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5. PROJECT OBJECTIVES

The main objectives of this project are:

1. Analyze overall sales and revenue performance.
2. Identify monthly and seasonal sales trends.
3. Evaluate product and category performance.
4. Calculate key e-commerce performance indicators.
5. Understand customer purchasing behavior.
6. Measure repeat customer activity.
7. Identify high-value customer segments.
8. Analyze order cancellations and product returns where data is available.
9. Develop an interactive Power BI dashboard.
10. Recommend data-driven strategies for business improvement.

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6. DATASET DESCRIPTION

For this project, a hypothetical transactional dataset named `ecommerce_sales.csv` is proposed.

The dataset would contain one record per order line, meaning an order containing multiple products could appear across multiple rows.

The actual dataset should be obtained from a legitimate public dataset or generated for demonstration purposes before the analysis is executed.

6.1 Proposed Dataset Columns

Column	Description
Order_ID	Unique order identifier
Order_Date	Date the order was placed
Customer_ID	Unique customer identifier
Product_ID	Unique product identifier
Product_Name	Name of the purchased product
Category	Product category
Quantity	Number of units purchased
Unit_Price	Price per unit
Discount	Discount applied to the order line
Payment_Method	Method used to pay
Order_Status	Order status
Shipping_Cost	Shipping cost associated with the order
Return_Status	Indicates whether the item was returned
Region	Customer's sales region

Additional fields, such as acquisition channel, advertising expenditure, product cost, delivery date, and customer registration date, could be included if the source dataset provides them.

6.2 Data Requirements

A useful dataset should contain enough information to calculate sales metrics and examine purchasing behavior.

The following checks should be performed:

- Verify that order identifiers and product identifiers are correctly formatted.
- Confirm that transaction dates can be interpreted as dates.
- Identify missing customer or product information.

- Investigate duplicate transaction records.
- Check for negative quantities or invalid prices.
- Determine how discounts and returns are represented.
- Verify whether cancelled orders are included.
- Confirm the currency and treatment of taxes and shipping charges.

These checks are important because incorrect assumptions can lead to inaccurate revenue calculations.

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7. KEY PERFORMANCE INDICATORS

Key Performance Indicators (KPIs) provide a consistent way to measure e-commerce performance.

7.1 Total Revenue

Total revenue represents the value of sales during a specified period.

A basic order-line calculation is:

Net Merchandise Sales = Quantity × Unit Price − Discount

The calculation must be adjusted if the source data already stores discounted line totals or uses a different discount convention.

Refunds, cancellations, taxes, and shipping charges should be handled consistently with the company's reporting policy.

7.2 Total Orders

Total Orders measures the number of distinct orders placed during a period.

Formula:

Total Orders = Count of Distinct Order IDs

Counting distinct orders prevents orders containing multiple products from being counted multiple times.

The treatment of cancelled or fully refunded orders should be clearly documented.

7.3 Average Order Value

Average Order Value (AOV) measures the average merchandise sales value per order.

Formula:

AOV = Net Merchandise Sales / Number of Eligible Orders

The revenue definition and eligible order population must be consistent across reports.

7.4 Total Units Sold

Total Units Sold measures the number of product units purchased.

Formula:

Total Units Sold = Sum of Eligible Transaction Quantities

Returns and cancellations may require adjustments depending on whether the metric measures gross sales or net fulfilled sales.

7.5 Repeat Purchase Rate

Repeat Purchase Rate measures the proportion of customers who have purchased more than once within a specified period.

Formula:

Repeat Purchase Rate = Customers With Multiple Purchases / Customers With at Least One Purchase × 100

The measurement period must be specified because repeat purchasing can change significantly across different time windows.

7.6 Customer Lifetime Value

Customer Lifetime Value (CLV) estimates the value associated with a customer over their relationship with the business.

A simple historical measure is the total net sales attributed to a customer over the selected observation period.

A predictive CLV model requires additional assumptions and data, such as purchase frequency, customer lifespan, margins, and retention behavior.

Revenue-based CLV should not be confused with profit-based customer value.

7.7 Return Rate

Return Rate measures the proportion of eligible purchased items that were returned.

Formula:

Return Rate = Returned Units / Eligible Units Sold × 100

A separate order-level return rate may also be calculated.

The denominator must be defined carefully, especially when the dataset includes partial returns, cancellations, or refunds.

7.8 Conversion Rate

Conversion Rate measures the proportion of eligible website sessions that result in a purchase.

Formula:

Conversion Rate = Converting Sessions / Eligible Sessions × 100

This metric cannot be calculated from order transactions alone. Website session or visitor data is required.

7.9 Customer Acquisition Cost

Customer Acquisition Cost (CAC) measures the cost of acquiring new customers.

Formula:

CAC = Eligible Acquisition and Marketing Costs / Number of New Customers Acquired

CAC requires marketing expenditure and customer acquisition data.

It should not be estimated from sales transactions alone.

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8. PROJECT METHODOLOGY

The project follows a structured data analytics workflow.

Step 1: Data Collection

Obtain a suitable e-commerce dataset from a legitimate source.

Import the dataset into Python or Excel for initial inspection.

Step 2: Data Understanding

Review column names, data types, dataset dimensions, and summary statistics.

Identify which fields are available for the planned KPIs.

Step 3: Data Cleaning

Handle missing values, investigate duplicate records, standardize categories, and correct data types.

Remove records only when justified by data quality rules.

Step 4: Data Transformation

Calculate line-level sales, derive order month and year, and create any additional fields needed for analysis.

Step 5: Exploratory Data Analysis

Examine revenue trends, product performance, customer behavior, and regional patterns.

Step 6: SQL Analysis

Write SQL queries to calculate KPIs, group transactions by month, rank products, and segment customers.

Step 7: Dashboard Development

Connect the cleaned dataset to Power BI and create interactive visualizations.

Step 8: Business Interpretation

Translate analytical findings into recommendations tied to measurable business objectives.

Step 9: Validation

Reconcile dashboard totals against the underlying dataset and verify that the KPI definitions are applied consistently.

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9. DATA CLEANING AND PREPROCESSING

Data cleaning is essential because transaction datasets may contain missing values, duplicate records, inconsistent categories, or invalid measurements.

9.1 Missing Values

Missing values should be examined column by column.

For example:

- Missing customer identifiers may prevent customer-level analysis.
- Missing product categories may affect category reporting.
- Missing transaction dates may prevent time-series analysis.
- Missing prices or quantities may prevent reliable sales calculations.

Missing values should not automatically be replaced with zero. The appropriate treatment depends on the meaning of the field.

9.2 Duplicate Records

Duplicate records should be investigated using transaction identifiers and the structure of the source dataset.

A repeated Order_ID does not necessarily indicate a duplicate because an order may contain multiple products.

9.3 Data Type Correction

Dates should be converted into a consistent date format.

Prices, quantities, discounts, and shipping costs should use appropriate numeric types.

Categorical columns should be standardized to prevent inconsistencies such as `Electronics`, `electronics`, and `ELECTRONICS` being treated as separate categories.

9.4 Invalid Values

Check for:

- Negative prices that are not legitimate adjustments.
- Zero or negative quantities that are not valid returns.
- Discounts outside the permitted range.
- Invalid dates.
- Inconsistent order statuses.
- Duplicate product or customer identifiers.

9.5 Revenue Calculation

Revenue calculations should reflect the structure of the dataset.

If `Discount` represents an absolute monetary discount:

Line Sales = Quantity × Unit Price − Discount

If `Discount` represents a percentage, the calculation must instead use the percentage discount.

The source documentation should be checked before applying either formula.

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10. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis helps reveal patterns and relationships in the transaction data.

10.1 Sales Trend Analysis

Monthly sales should be calculated and plotted over time.

A line chart can help identify:

- Growth or decline in sales.
- Seasonal fluctuations.
- Periods of unusually high demand.
- Changes following promotional campaigns.

However, seasonality should not be inferred from a single short observation period.

10.2 Product Performance Analysis

Products can be ranked by:

- Net sales revenue.
- Units sold.
- Number of distinct orders containing the product.
- Return rate.
- Profit, if cost information is available.

This helps distinguish high-volume products from high-revenue or high-margin products.

10.3 Category Analysis

Revenue and units sold can be aggregated by product category.

A bar chart can show which categories contribute the most sales.

Category revenue should not automatically be interpreted as category profitability because product costs and operational expenses may differ.

10.4 Regional Analysis

If regional information is available, revenue and order volume can be compared across regions.

This can reveal geographic differences in sales performance and help identify regions requiring further investigation.

10.5 Customer Purchasing Behavior

Customer-level analysis can examine:

- Total orders per customer.
- Total sales per customer.
- Average order value.
- Time between purchases.
- Repeat purchase behavior.

Customer segmentation can then be used to compare purchasing patterns.

10.6 Return Analysis

If return information is available, return rates can be compared across products and categories.

A high return rate may indicate product quality issues, inaccurate product descriptions, sizing problems, or customer expectation mismatches.

These explanations should be investigated rather than assumed.

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11. SQL ANALYSIS

SQL can be used to extract and aggregate information from the transactional database.

The following examples assume a table named `ecommerce_sales` containing the proposed fields.

The SQL syntax may need minor adjustments depending on the database platform.

11.1 Calculate Total Merchandise Sales

```
SELECT
    SUM(Quantity * Unit_Price - Discount) AS Total_Sales
FROM ecommerce_sales
WHERE Order_Status = 'Completed';
```

This example assumes that `Discount` is an absolute monetary amount and that completed orders represent the desired reporting population.

The treatment of returns and refunds must be incorporated if the objective is to calculate net sales after returns.

11.2 Calculate Total Orders

```
SELECT
    COUNT(DISTINCT Order_ID) AS Total_Orders
FROM ecommerce_sales
WHERE Order_Status = 'Completed';
```

This query counts distinct completed orders rather than transaction lines.

11.3 Calculate Monthly Sales

```
SELECT
    EXTRACT(YEAR FROM Order_Date) AS Sales_Year,
    EXTRACT(MONTH FROM Order_Date) AS Sales_Month,
    SUM(Quantity * Unit_Price - Discount) AS Monthly_Sales
FROM ecommerce_sales
WHERE Order_Status = 'Completed'
GROUP BY
    EXTRACT(YEAR FROM Order_Date),
    EXTRACT(MONTH FROM Order_Date)
ORDER BY
    Sales_Year,
    Sales_Month;
```

Date extraction syntax varies between database platforms.

11.4 Identify Top-Selling Products

```
SELECT
    Product_ID,
    Product_Name,
    SUM(Quantity) AS Units_Sold,
    SUM(Quantity * Unit_Price - Discount) AS Product_Sales
FROM ecommerce_sales
WHERE Order_Status = 'Completed'
GROUP BY
    Product_ID,
    Product_Name
ORDER BY
    Product_Sales DESC
LIMIT 10;
```

This query identifies the ten products with the highest calculated sales.

11.5 Identify Repeat Customers

```
SELECT
    Customer_ID,
    COUNT(DISTINCT Order_ID) AS Number_Of_Orders
FROM ecommerce_sales
WHERE Order_Status = 'Completed'
GROUP BY Customer_ID
HAVING COUNT(DISTINCT Order_ID) > 1
ORDER BY Number_Of_Orders DESC;
```

This query identifies customers with more than one completed order.

A repeat purchase rate can then be calculated using the count of repeat customers divided by the total number of eligible customers.

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12. POWER BI DASHBOARD DESIGN

An interactive Power BI dashboard can bring the main analytical findings together in one interface.

The proposed dashboard contains four pages.

Page 1: Executive Overview

KPI Cards

- Total Net Sales
- Total Orders
- Average Order Value
- Total Units Sold
- Repeat Purchase Rate

Visualizations

- Monthly sales trend.
- Revenue by product category.
- Sales by region.
- Top ten products by revenue.

Purpose

Provide management with a quick overview of overall business performance.

Page 2: Product Performance

Visualizations

- Revenue by product.
- Units sold by product.
- Revenue by category.
- Return rate by category.
- Product sales ranking.

Purpose

Identify strong-performing products and categories requiring further investigation.

Page 3: Customer Insights

Visualizations

- New versus repeat customers.
- Orders per customer.
- Customer revenue distribution.
- Repeat purchase rate.
- Recency, frequency, and monetary segmentation.

Purpose

Understand customer purchasing behavior and identify opportunities for retention campaigns.

Page 4: Regional and Operational Analysis

Visualizations

- Revenue by region.
- Orders by region.
- Order status distribution.
- Returns by region.
- Delivery performance, if delivery dates are available.

Purpose

Understand geographic sales patterns and operational performance.

Interactive Filters

The dashboard can include filters for:

- Date range.
- Product category.
- Region.
- Order status.
- Payment method.

These filters allow users to investigate specific business questions without creating separate reports.

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13. CUSTOMER SEGMENTATION

Customer segmentation groups customers according to their purchasing behavior.

One suitable approach is RFM analysis.

RFM stands for:

- **Recency:** How recently the customer purchased.
- **Frequency:** How often the customer purchased.
- **Monetary Value:** How much the customer spent.

13.1 Recency

Recency measures the number of days since a customer's most recent purchase, relative to a defined analysis date.

A smaller recency value indicates a more recent purchase.

13.2 Frequency

Frequency measures the number of distinct eligible orders placed by a customer during the selected observation period.

13.3 Monetary Value

Monetary value measures the customer's total eligible sales over the observation period.

If profit data is available, customer contribution margin may provide a more useful measure than revenue alone.

13.4 Example Customer Segments

Customer Segment	Characteristics	Possible Strategy
High-Value Customers	High spending and frequent purchases	Loyalty benefits and personalized recommendations
Loyal Customers	Frequent purchases	Rewards and early access to products
Recent Customers	Purchased recently but have limited history	Welcome offers and product recommendations
At-Risk Customers	Previously active but have not purchased recently	Targeted re-engagement campaigns

Customer Segment	Characteristics	Possible Strategy
Low-Activity Customers	Low purchase frequency and spending	Relevant promotions and product discovery

These segments should be defined using actual data and clearly documented thresholds.

A customer's absence of recent purchases does not necessarily mean they have disengaged; purchasing cycles vary by product category.

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14. BUSINESS INSIGHTS AND INTERPRETATION

Because this is a hypothetical case study, the following are potential findings to investigate rather than measured results.

14.1 Sales Performance

Question: Is revenue growing or declining over time?

Monthly sales analysis can reveal whether revenue is increasing, decreasing, or fluctuating.

Business application: Management can investigate unusual changes and adjust marketing or inventory plans accordingly.

14.2 Product Performance

Question: Which products generate the highest revenue?

Ranking products by revenue and units sold can reveal differences between product popularity and financial contribution.

Business application: The company can prioritize product availability and evaluate merchandising opportunities.

14.3 Customer Retention

Question: How many customers make repeat purchases?

Customer-level analysis can identify the proportion of customers who purchase more than once.

Business application: The company can evaluate loyalty programs, re-engagement campaigns, and personalized recommendations.

14.4 Average Order Value

Question: How much does the average order contribute to sales?

Average Order Value can be tracked over time and compared across customer segments or product categories.

Business application: The company can test product bundles, cross-selling, and free-shipping thresholds.

14.5 Returns

Question: Which products or categories experience unusually high return rates?

Return analysis can help identify potential quality or product-description issues.

Business application: The company can investigate product listings, sizing guidance, packaging, and fulfillment processes.

14.6 Regional Performance

Question: Which regions generate the greatest revenue?

Regional analysis can reveal geographic differences in demand.

Business application: The company can evaluate regional marketing, distribution, and delivery strategies.

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15. BUSINESS RECOMMENDATIONS

The recommendations below are proposed actions. Their effectiveness should be measured through actual business data and controlled experiments where appropriate.

15.1 Improve Customer Retention

Use purchase history to identify customers who may benefit from relevant reminders, personalized recommendations, or loyalty incentives.

Measure results using repeat purchase rate, retention cohorts, and incremental revenue.

15.2 Optimize Product Assortment

Combine revenue, sales volume, return rates, and product margin to evaluate product performance.

Avoid relying on revenue rankings alone when deciding which products to prioritize.

15.3 Improve Inventory Planning

Use historical demand patterns to inform inventory planning.

For products with seasonal demand, consider historical sales patterns and lead times.

Inventory decisions should also account for stock availability, supplier constraints, and forecast uncertainty.

15.4 Strengthen Marketing Measurement

Connect campaign expenditure and customer acquisition data to sales records.

Measure acquisition cost and incremental outcomes rather than judging campaign effectiveness solely by clicks or total revenue.

15.5 Reduce Product Returns

Investigate products with unusually high return rates.

Potential actions include improving product descriptions, providing more accurate product images, and making sizing or compatibility information clearer.

15.6 Increase Average Order Value

Test product bundles, complementary recommendations, and free-shipping thresholds.

Evaluate whether these changes increase contribution margin rather than simply increasing order value.

15.7 Establish Consistent Reporting

Create standardized KPI definitions and a centralized dashboard.

This reduces discrepancies between reports and allows different teams to work from consistent metrics.

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16. EXPECTED BUSINESS BENEFITS

If implemented effectively and validated using real business data, the proposed analytical framework could help the business:

1. Improve visibility into sales performance.
2. Identify high-performing products and categories.
3. Understand customer purchasing patterns.
4. Support customer retention initiatives.
5. Improve inventory planning.
6. Investigate product return patterns.
7. Measure marketing effectiveness more accurately.
8. Standardize business reporting.
9. Support evidence-based decision-making.
10. Identify opportunities for sustainable growth.

These are potential benefits, not measured outcomes of the hypothetical analysis.

Actual impact would depend on data quality, implementation, business conditions, and the effectiveness of the resulting decisions.

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17. LIMITATIONS

Several limitations should be considered.

17.1 Hypothetical Business Scenario

ShopSphere is an illustrative company, and no real company's data has been analyzed.

The findings must not be presented as measured company performance.

17.2 Dataset Availability

The proposed dataset structure is a specification, not an analyzed dataset.

Actual sales figures, trends, and customer insights cannot be established until a dataset is obtained and processed.

17.3 Limited Customer Information

Transaction records alone may not explain why customers purchase or stop purchasing.

Additional information, such as marketing interactions and customer feedback, could improve the analysis.

17.4 Missing Cost Data

Revenue does not equal profit.

Without product costs, fulfillment expenses, payment fees, and marketing costs, profitability cannot be measured reliably.

17.5 Limited Marketing Data

Conversion rate and customer acquisition cost require additional information beyond order transactions.

17.6 Correlation Versus Causation

Observed relationships do not prove that a business action caused a change in sales or customer behavior.

Controlled experiments or appropriate causal analysis may be necessary to evaluate interventions.

17.7 Data Quality

Missing values, incorrect transaction records, inconsistent return handling, and duplicate data can affect the reliability of the results.

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18. FUTURE SCOPE

The project can be extended in several directions.

18.1 Sales Forecasting

Time-series forecasting models could be developed to estimate future sales using historical transaction data.

18.2 Customer Churn Analysis

Customer activity data could be used to identify patterns associated with declining engagement.

Any churn definition should reflect the purchasing cycle of the business.

18.3 Recommendation Systems

Product interaction and purchase history could be used to develop personalized product recommendations.

18.4 Market Basket Analysis

Association-rule mining could identify products frequently purchased together.

These insights could support product bundling and cross-selling.

18.5 Profitability Analysis

Adding product costs, marketing costs, and fulfillment expenses would allow analysis of gross margin and contribution margin.

18.6 Cohort Analysis

Customers could be grouped by their first purchase month to compare retention and purchasing behavior over time.

18.7 Automated Reporting

Scheduled data refreshes and automated KPI reports could reduce manual reporting work and improve operational visibility.

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19. CONCLUSION

This case study presents a proposed data analytics framework for understanding e-commerce sales performance and customer behavior.

The hypothetical business scenario highlights common challenges involving revenue visibility, product performance, customer retention, inventory planning, and reporting consistency.

The proposed solution combines Python for data cleaning and exploratory analysis, SQL for transactional analysis, and Power BI for dashboard development.

Key metrics include total sales, order volume, average order value, repeat purchase rate, customer lifetime value, and return rate.

Customer segmentation and product analysis can provide additional insights into purchasing behavior and business opportunities.

The project demonstrates how a structured data analytics workflow can support better business decisions.

However, actual business findings and measurable improvements can only be established after obtaining, validating, and analyzing real or clearly identified demonstration data.

The next step is to implement the workflow using a real public e-commerce dataset and document the resulting metrics, visualizations, and findings.

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20. TECHNOLOGIES USED

Programming Language: Python

Data Analysis Libraries:

- Pandas
- NumPy
- Matplotlib
- Seaborn

Database Query Language: SQL

Business Intelligence Tool: Microsoft Power BI

Core Analytical Techniques:

- Data Cleaning
- Exploratory Data Analysis
- KPI Development
- Sales Trend Analysis
- Customer Segmentation
- Product Performance Analysis
- Dashboard Development

Project Type: Business Intelligence and Data Analytics

Dataset: Proposed e-commerce transactional dataset

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21. PROJECT DISCLAIMER

This case study uses a hypothetical e-commerce business scenario.

The proposed dataset schema, analytical workflow, business questions, and recommendations are illustrative.

No actual company data has been analyzed, and no numerical business improvements or findings are claimed.

Any results added to a final portfolio version should be calculated from a real dataset or clearly identified synthetic data.

22. LINKEDIN PROJECT SUMMARY

E-Commerce Sales Analysis and Customer Insights | Python, SQL & Power BI

I explored how data analytics can help an e-commerce business understand sales performance, customer purchasing behavior, and product trends.

The case study proposes an end-to-end analytics workflow covering data cleaning, exploratory data analysis, SQL queries, KPI development, customer segmentation, and interactive Power BI dashboard design.

The key business questions include:

- Which products and categories generate the most revenue?
- How do sales change over time?
- What proportion of customers make repeat purchases?
- How can product returns be investigated?
- How can dashboards support better business decisions?

Tools and technologies: Python, Pandas, NumPy, SQL, Excel, and Power BI.

This project helped me explore the relationship between data analysis and business decision-making, with a focus on turning transactional data into actionable insights.

Note: This is a hypothetical case study. Numerical results and business improvements will be reported only after analysis of an actual dataset.

#DataAnalytics #Ecommerce #Python #SQL #PowerBI #BusinessIntelligence #DataVisualization #CustomerAnalytics